

GHANA INSTITUTE OF MANAGEMENT AND PUBLIC ADMINISTRATION

School of Technology | Department of Computer Science

CS 305: Problem Set 3 (Turing Machines & Decidability)

Weeks 9–11 | Total Marks: 100 | Weight: 7.5% of Final Grade

Instructions & Guidelines

- Turing machine transition functions must follow the format $\delta(q, a) = (r, b, L/R)$.
- High-level implementation descriptions (Sipser Stage 1, Stage 2...) must be precise and account for tape boundaries.
- Decidability proofs must explicitly construct a decider machine that is guaranteed to halt on all inputs.

Problem 1: Low-Level Turing Machine Design (25 Marks)

Let $\Sigma = \{0, 1\}$. Design a single-tape Deterministic Turing Machine $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$ that decides the non-context-free language:

$$L_1 = \{0^n 1^n 0^n \mid n \geq 0\}$$

- State the formal components: state set Q , tape alphabet Γ , start state q_0 , q_{accept} , and q_{reject} . (10 Marks)
- Provide the complete transition function table for δ , and trace the configuration sequence of M on the input string $w = 010$. (15 Marks)

Problem 2: Multi-Tape to Single-Tape Equivalence (25 Marks)

Let M be a k -tape Turing machine that runs in time $t(n)$ on inputs of length n .

- Describe the formal simulation algorithm by which a single-tape Turing machine S simulates the execution of k -tape machine M (Sipser Theorem 3.13). (15 Marks)
- Rigorously prove that the total running time of the single-tape simulator S is $\mathcal{O}((t(n))^2)$. (10 Marks)

Problem 3: Decidable Language Decision Procedures (25 Marks)

Prove that the following languages are ****Turing-decidable**** by constructing explicit deciders that are guaranteed to halt on every input:

- $E_{\text{DFA}} = \{\langle A \rangle \mid A \text{ is a DFA and } L(A) = \emptyset\}$. (12 Marks)
- $EQ_{\text{DFA}} = \{\langle A, B \rangle \mid A, B \text{ are DFAs and } L(A) = L(B)\}$. (Hint: Use the symmetric difference construction $L(C) = (L(A) \cap \overline{L(B)}) \cup (\overline{L(A)} \cap L(B))$). (13 Marks)

Problem 4: Modern Applied Case Study — Chain-of-Thought in Large Language Models (25 Marks)

In modern theoretical machine learning, fixed-depth autoregressive Transformers are known to have limited computational depth (bounded circuit class TC^0). However, when augmented with **Chain-of-Thought** (scratchpad memory), their computational expressivity expands dramatically.

- (a) Formulate how an autoregressive LLM outputting intermediate tokens on an external scratchpad simulates the read/write tape transitions of a Universal Turing Machine. (12 Marks)
- (b) Explain why the Church–Turing Thesis implies that no neural network architecture, regardless of parameter count or dataset size, can exceed the computational recognition capability of a Turing Machine. (13 Marks)